QUALITY INFORMATION DOCUMENT

For
OMI_HEALTH_TEMPSALOXY_BALTIC_cod_volume

Issue: 4.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/10/10	all	Creation of the document	U. Raudsepp I. Maljutenko	P. Lagemaa
1.1	2020/04/03		Minor corrections from MOI taken into account	P. Lagemaa	V. Huess
2.1	2021/10/01	all	Remove temporal extension specific details	U. Raudsepp I. Maljutenko	P. Lagemaa
3.0	2023/09/01	all	Update the template and minor changes	U. Raudsepp I. Maljutenko	P. Lagemaa
4.0	2025/05/22	all	Template change+ minor changes	P. Lagemaa	V. Huess

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1) INTRODUCTION

a) Product Name

OMI_HEALTH_TEMPSALOXY_BALTIC_cod_volume.

b) Indicator Evaluation Method

The cod reproductive volume is computed from salinity, temperature and oxygen concentrations as described by Raudsepp et al. (2018) and as previously published in the literature by Plikshs et al (2015), for instance. The daily salinity and temperature data is extracted from the BALTICSEA_MULTIYEAR_PHY_003_011 product and oxygen concentrations have been extracted from the BALTICSEA_MULTIYEAR_BGC_003_012 products. The accuracy of the index is directly dependent on the accuracy of the Baltic reanalysis products (BALTICSEA_MULTIYEAR_PHY_003_011 and BALTICSEA_MULTIYEAR_BGC_003_012) therefore evaluation is based on the QUID documents of the products.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf
BALTICSEA_MULTIYEAR_BGC_003_012 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-012.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf

2) SCIENTIFIC VALIDATION

a) Cod Reproductive Volume

The cod reproductive volume index is calculated based on modelled salinity, temperature and oxygen data, hence the accuracy of the BALTIC_OMI_HEALTH_codt_volume depends on these input variables.

In the CMEMS-BAL-QUID-003-011 the salinity of reanalysis product has been compared with all available measurements across the Baltic for period of 1993 until one year to present. As the data assimilation is employed the model results of reanalysis show only small biases (<0.2 g kg-1) compared to observations at all observed depth ranges. The RMSE showed highest values around 0.8 g kg-1 in the subsurface layer of 30-80 m and decreased toward the deeper layers to approximately 0.25 g kg-1 in the layer of 80-200 m.

The modelled temperature bias showed high temporal variability especially at surface where deviations from observations were in range of ± 0.2 °C. The average bias remained within range ± 0.06 °C at all depths.

The comparison of simulated oxygen data with the observations from different independent sources showed that the modelled oxygen deviations from observations are in most cases much smaller than the observed standard deviations. The subsurface deviations of oxygen in the layer of 80-200 m remained between 60 - 90 mmol m-3 (1.36 – 2.05 ml l-1) and the RMSE <93 mmol m-3. As the reanalysis product has large oxygen bias in the subsurface layer the oxygen values were bias corrected before calculations of cod reproductive volumes (Figure 1).

Scientific evaluation can be found in the Copernicus Marine Service Ocean State Report (Raudsepp et al., 2018).

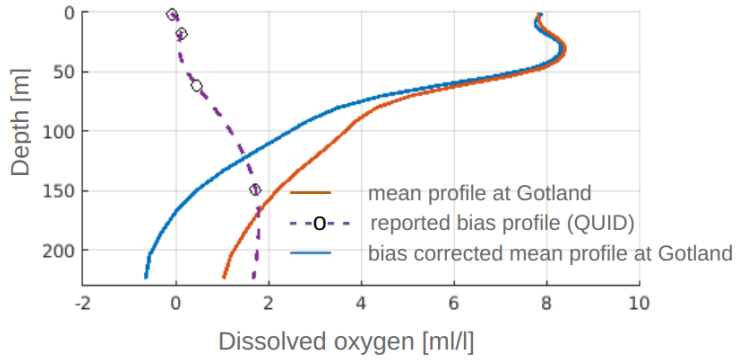

Figure 1. Time-averaged profiles of dissolved oxygen in the central Gotland Basin from the Baltic Sea reanalysis. (BALTICSEA_MULTIYEAR_BGC_003_012).

3) REFERENCES

Raudsepp, U., Maljutenko, I., Kõuts, M., 2019. Cod reproductive volume potential in the Baltic Sea. In: Copernicus Marine Service Ocean State Report, Issue 3, Journal of Operational Oceanography, 12:sup1, s26–s30; DOI: 10.1080/ 1755876X.2019.1633075

Plikshs M, Hinrichsen H-H, Elferts D, Sics I, Kornilovs G, Köster FW. 2015. Reproduction of Baltic cod, Gadus morhua (Actinopterygii: Gadiformes: Gadidae), in the Gotland basin: causes of annual variability. Acta Ichthyol Piscat. 45 (3):247–258. doi:10.3750/AIP2015.45.3.04